

ONE-WAVE FRAMEWORK

Book 1 — Micro

Chapter 11: No Antimatter — Internal Pressure, Not Mirror Particles

Version: 3.0 Date: July 1, 2026 Class: B — Applied Layer Spine: Gray / 2D / 3D / Mathematics / Predictions / Yellow Audit / Future Work / Closing Thoughts

Dependencies: B-03 Expression, B-04 Compression, B-06b Four Views, B-01 Balance, E-03 Surface, B-09 Break Condition, F-05 Interference Status: GREEN (claim) / YELLOW (formal derivation)

Gray — Standard Model Reference

Antimatter consists of antiparticles — particles with the same mass but opposite quantum numbers. The positron is the antiparticle of the electron: same mass ($0.511 \text{ MeV}/c^2$), opposite charge (+1). The antiproton is the antiparticle of the proton: same mass, charge -1. Dirac predicted antimatter in 1928 from his relativistic quantum equation. When matter meets antimatter, annihilation occurs: both convert to photons. $e^+ + e^- \rightarrow 2 \text{ gamma}$ (511 keV photons)

The antimatter asymmetry problem: The Big Bang should have produced equal amounts of matter and antimatter. If so, they would have annihilated completely. Nothing would exist. But the universe exists. There is far more matter than antimatter. This matter-antimatter asymmetry is one of the deepest unsolved problems in physics. CP violation (charge-parity symmetry violation) is measured but insufficient to explain the observed asymmetry.

Standard Model strength: antimatter exists and is well described by QFT. Antihydrogen has been produced and measured at CERN. Standard Model limitation: the antimatter asymmetry problem is unsolved. Why the universe has matter at all is unexplained.

2D One-Wave Interpretation

In 2D, there is no antimatter.

There are modes with outward pressure (expression dominant). There are modes with inward pressure (compression dominant). These are not mirror particles of each other. They are the two phases of the same oscillation.

What we call “matter” is expression-dominant modes. What we call “antimatter” is the internal compression pressure that is released when an expression-dominant mode’s boundary breaks.

In 2D: the proton is a braided loop with expression dominant. Its interior is under outward pressure. When the boundary breaks, that outward pressure is released. That released pressure is the positron — not a separate particle. It is the proton’s own interior expressing outward when the container breaks.

3D One-Wave Interpretation

In 3D, antimatter is a misidentification of boundary pressure release events.

The positron: When a proton boundary breaks (B-09 Break Condition), the internal expression pressure (B-03) is released outward. That released pressure propagates as a mode with: - Positive pressure (expression dominant) - Spin-half (from the roll-off at the broken boundary, B-06b) - Opposite charge orientation to the electron (because the proton interior was positive)

This released pressure mode is what standard physics calls a positron. It is not a mirror particle that was created along with the electron. It is the proton's own interior pressure, released.

The electron (from Chapter 4) is the negative pressure cushion at the electron shell boundary — the outward-pointing pressure gradient. The positron is the positive pressure release from a broken proton boundary.

They annihilate when they meet because: A positive pressure mode meeting a negative pressure mode produces cancellation. F-05 Interference with $\phi = \pi \Rightarrow \psi_T = 0$. The two pressure modes cancel each other. The energy goes into photons (pressure oscillations, Chapter 7).

The antimatter asymmetry problem dissolved: The Big Bang did not produce equal matter and antimatter. It produced expression-dominant modes (matter). The compression pressure that accompanies them is not a separate equal quantity of antimatter. It is the internal pressure of the matter modes themselves. When matter modes form, their compression pressure is internal, not external. There was never a separate equal quantity of antimatter to annihilate with.

The universe has matter and no antimatter because antimatter is not a separate substance — it is a boundary pressure event.

Pair production: When a high-energy photon passes near a nucleus, it can produce an electron-positron pair (e+e- pair production). In One-Wave: the photon's pressure oscillation, focused by the nuclear field gradient, creates a local region of excess pressure. This pressure splits into two modes: - One mode with positive boundary-pressure sign (the positron) - One mode with negative boundary-pressure sign (the electron) These labels specify signed boundary pressure, not spatial gradient direction or whole-mode Compression/Expression dominance. These are not created from nothing. They are the photon's pressure being divided into two complementary boundary modes by the nuclear field.

Mathematics

Positron as released internal pressure (B-06b): $\Delta E = E_{\text{mode}} - E_s$ $\Delta E > 0$ at boundary break $\Rightarrow$ internal expression pressure released

Released pressure mode: $P_{\text{released}} = P_{\text{internal}}(\text{proton}) - P_{\text{boundary}}(\text{surface tension})$ $P_{\text{released}} > 0 \Rightarrow$ outward propagating mode $\Rightarrow$ positron

Annihilation as destructive interference (F-05): e+ pressure: $+P_c$ positive boundary-pressure sign; outward-propagating release e- pressure: $-P_c$ negative boundary-pressure sign; outward-pointing shell-boundary gradient Combined signed pressure: $\psi_T = +P_c + (-P_c) = 0$

Formal distinction: $P_q = s_q * P_c$, where s_q in $\{-1, +1\}$ $G_q = \text{grad}(P_q)$ The sign s_q and the spatial direction of G_q are separate quantities. Energy: $E_{\text{annihilation}} = 2 * E_{\text{mode}} = 2 * m_e * c^2 = 2 * 0.511 \text{ MeV}$ Released as two photons: $E_{\text{gamma}} = 0.511 \text{ MeV}$ each (conservation of energy and momentum)

Pair production (sketch — Yellow): Photon energy: $E_{\text{photon}} \geq 2 * m_e * c^2 = 1.022 \text{ MeV}$ (threshold) In One-Wave: photon pressure oscillation $\geq$ threshold to split into two boundary modes $E_{\text{photon}} = E_{e+} + E_{e-}$ Both modes carry spin-half from boundary roll-off (B-06b).

Antimatter asymmetry: Standard Model: $N_{\text{matter}} - N_{\text{antimatter}} \gg 0$ (unexplained asymmetry) One-Wave: $N_{\text{antimatter}} = 0$ at formation. Positrons and antiprotons are produced only by boundary break events, not at Big Bang. The apparent antimatter asymmetry problem does not exist in One-Wave.

Correction Record — Charge Sign vs. Gradient Direction

An earlier version of this chapter labeled the electron “compression-dominant, inward” in the pair-production and annihilation sections. That wording contradicted this chapter’s own earlier electron definition and Book 1 Chapter 4. The local labels were corrected to distinguish signed boundary pressure from spatial gradient direction. See C-316 for the full audit record.

Predictions

1. The positron is the released internal pressure of a broken proton boundary. Prediction: positron production always accompanies proton boundary break events. Pair production (photon $\rightarrow e^+e^-$) is a photon pressure splitting into two complementary modes. No independent creation of antimatter from “nothing.”
 2. Annihilation is destructive interference between complementary pressure modes. Prediction: e^+e^- annihilation always produces exactly two photons at 0.511 MeV each (for low-energy annihilation) because the pressure cancellation releases energy as two equal and opposite photons to conserve momentum.
 3. The antimatter asymmetry problem does not exist in One-Wave. Prediction: no symmetric antimatter was produced at the Big Bang (Snap and Fill event). The universe’s matter dominance is not an asymmetry — it is the natural state. Expression-dominant modes (matter) form first. Their internal compression pressure is internal, not a separate antimatter.
 4. Antihydrogen has identical spectral lines to hydrogen. Prediction: an antihydrogen atom (antiproton + positron) has the same harmonic shell structure (D-05) as hydrogen because the shell condition depends on the pressure magnitude, not sign. This matches the CERN antihydrogen measurements.
 5. CP violation is a boundary effect, not a fundamental asymmetry. Prediction: CP violation in meson decays comes from slight asymmetries in the boundary break conditions (B-09) between matter and antimatter mode types. The asymmetry is small because the boundary conditions are nearly symmetric.
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Yellow Audit

- Pair production derivation from photon pressure splitting not yet formalized
 - CP violation from boundary asymmetry not yet derived
 - Formal proof that positron = released internal pressure not yet completed
 - Antihydrogen spectral equivalence derivation deferred
 - Antimatter asymmetry resolution requires full Snap and Fill cosmological model
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Future Work

Formalize positron as released internal pressure from B-06b. Derive pair production threshold from photon pressure splitting model. Compute CP violation magnitude from boundary break asymmetry. Derive antihydrogen spectral equivalence from harmonic shell condition. Connect antimatter asymmetry resolution to Snap and Fill cosmological model (Book 5).

Closing Thoughts

The antimatter asymmetry problem is one of the great mysteries of cosmology. If the Big Bang produced equal matter and antimatter, where did all the antimatter go?

In One-Wave, this is not a problem. It is a misframing. The Big Bang (Snap and Fill) did not produce antimatter. It produced expression-dominant modes — matter. The compression pressure that balances them is internal to each mode.

Antimatter is not a separate substance hiding somewhere in the universe. Antimatter is what you get when a matter mode breaks open and its interior pours out.

The universe does not have a matter-antimatter imbalance. The universe has matter. Period. The positrons and antiprotons we produce in accelerators are not remnants of Big Bang antimatter. They are pressure release events from high-energy collisions.

The mystery dissolves when you stop treating the positron as a mirror particle of the electron and start treating it as the electron shell's interior, released by breaking its container.

END OF BOOK 1 CHAPTER 11 One wave. Mirror builds. Mark Wright. Kitty Hawk V0.